Similarities Between Multisensory Integration of PPPD Patients and Controls

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Background

PPPD

Functional Neurological Disorder

Chronic underlying **dizziness** and balance impairment

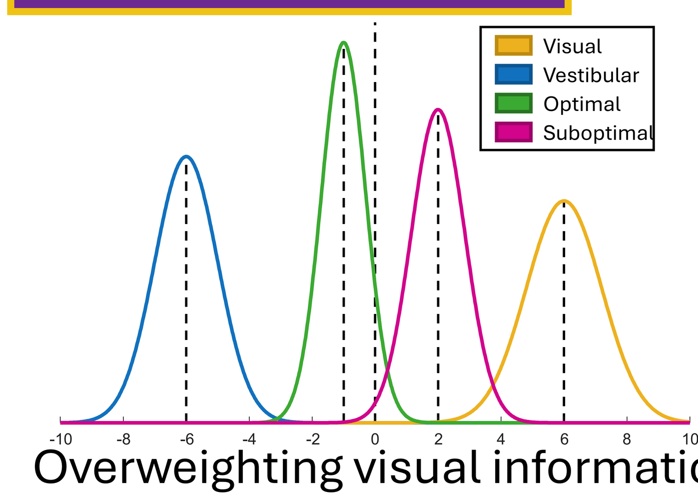
Severe bouts of dizziness caused by:

- Upright posture
- Self-movement
- Stimulating visual environments

Larger visually-induced balance distortion (e.g., 1,2)

Decreased activity in brain regions for **vestibular integration** (e.g., 3)

Overweighting Visual



Overweighting visual information can make integration suboptimal

Multisensory Integration

When two sources of information from different senses are **redundant**, we **integrate** them together to increase perceptual precision

This is effectively a **weighted sum** where the more precise the signal, the more its weight. Optimal weighting follows **Maximum Likelihood Estimation** (4), giving the best bias to precision trade-off

There is previous evidence for optimal visual and vestibular integration in healthy controls (5,6)

We measure multisensory integration with the tasks from papers 5&6.

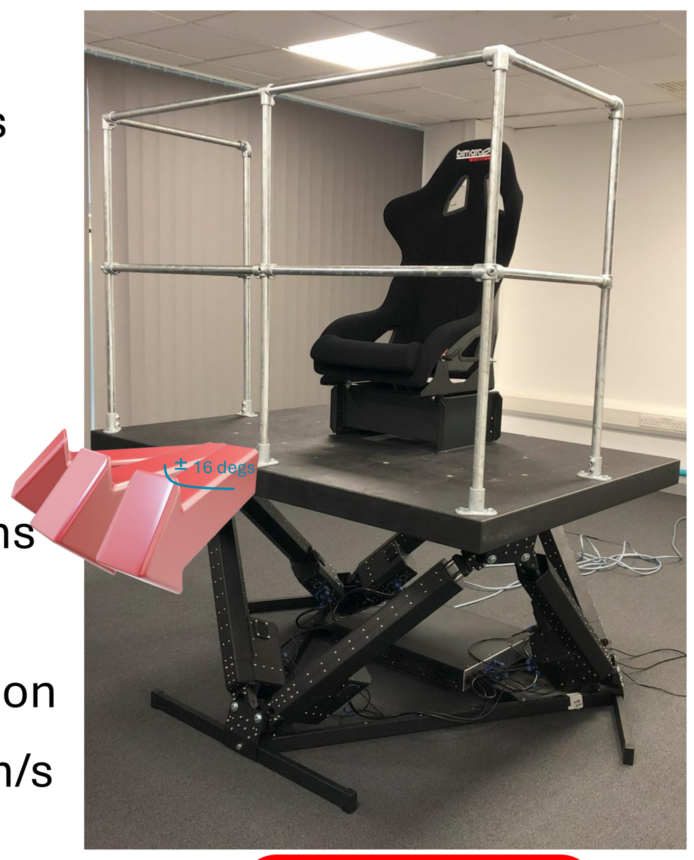
Integration Hypothesis:
*PPPD patients may **overweight** visual information during multisensory integration*

Design

Participants

31 PPPD patients and 25 age-matched controls (one control removed from SROM analysis)

Heading Task



Participants moved **virtually** in VR headset, **physically** on motion platform (right) or **both**

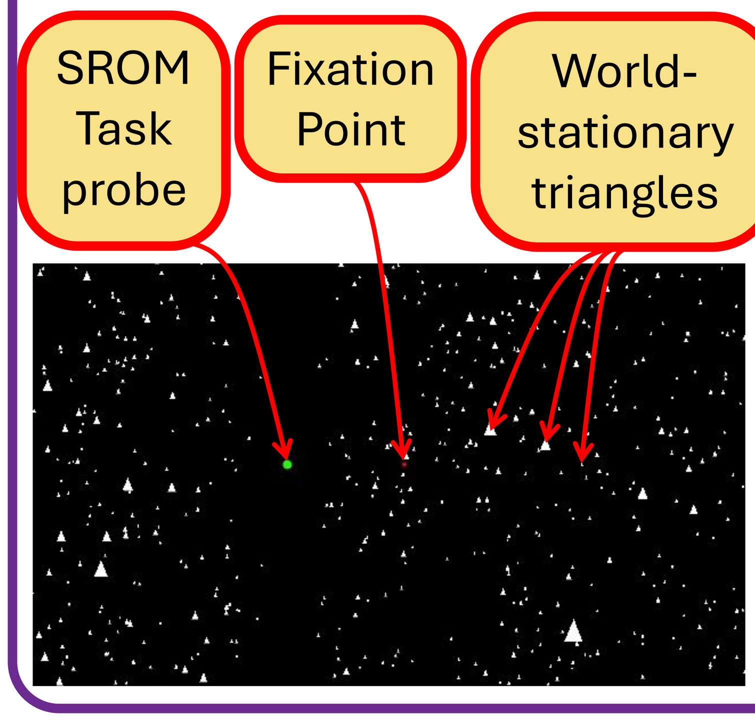
Creates unisensory and multisensory conditions

Max deviation from straight = 16degs

Adaptive Kesten staircase procedure on deviation

Gaussian acceleration curve, max speed 0.31m/s

Decision – **which side did you move?**



SROM Task

Scene-Relative Object Movement

Same design as the heading task but with a **vertically moving probe** in each condition

Participant moving either 16degs right or left

Adaptive Kesten staircase on probe motion

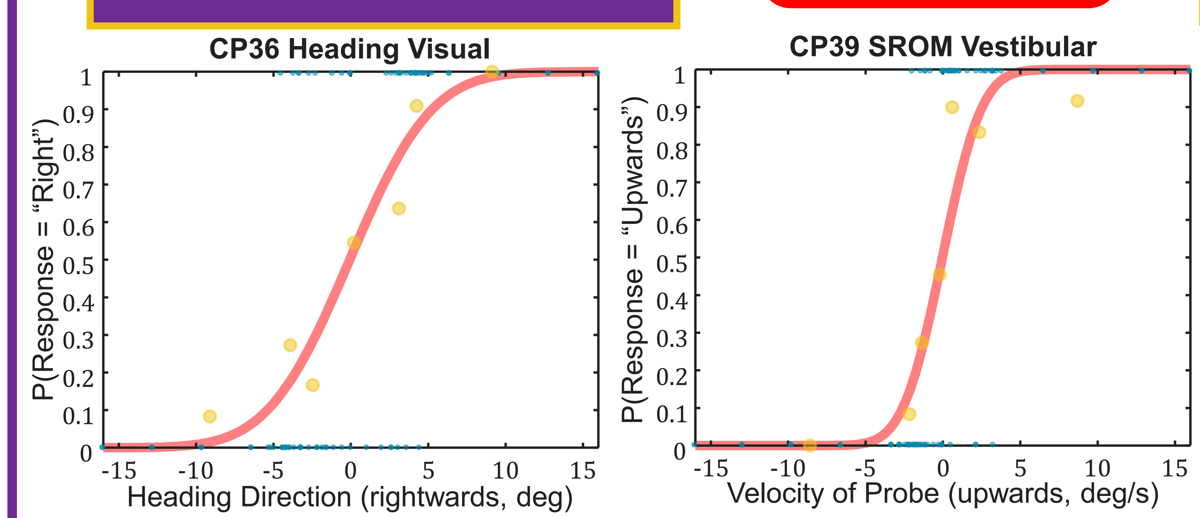
Decision – **did the probe move up or down?**

Visual weighting measured in Heading task
Two multisensory versions with 6deg discrepancy between visual and vestibular info

Results

Psychometric Functions

Same examples as in Haynes et al. (7)



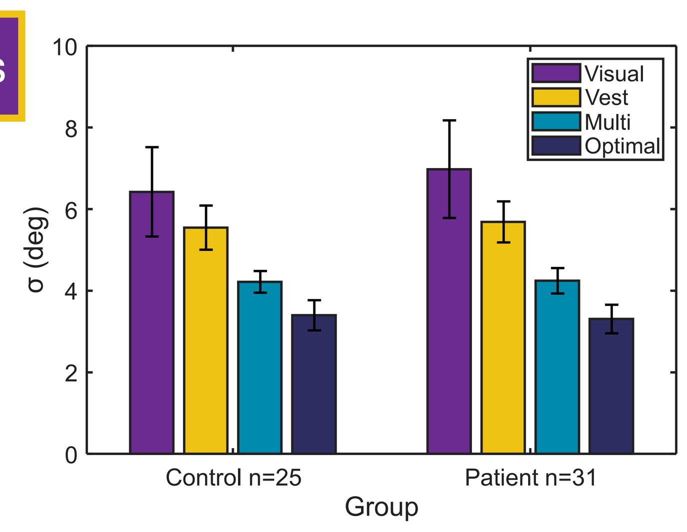
Using **behavioural responses** to the stimuli, we plot psychometric functions for the **probability** of a rightwards or upwards response based on **any heading direction or probe velocity**

We recover the **PSE** (stimulus value at P(0.5)) and **threshold** (stimulus value at P(0.841) – PSE)

PSE and threshold recovered for unisensory and multi-sensory conditions and **compared** to MLE **predictions**

We discuss the differences between the Heading and SROM tasks in our preprint paper (7)

Heading Thresholds

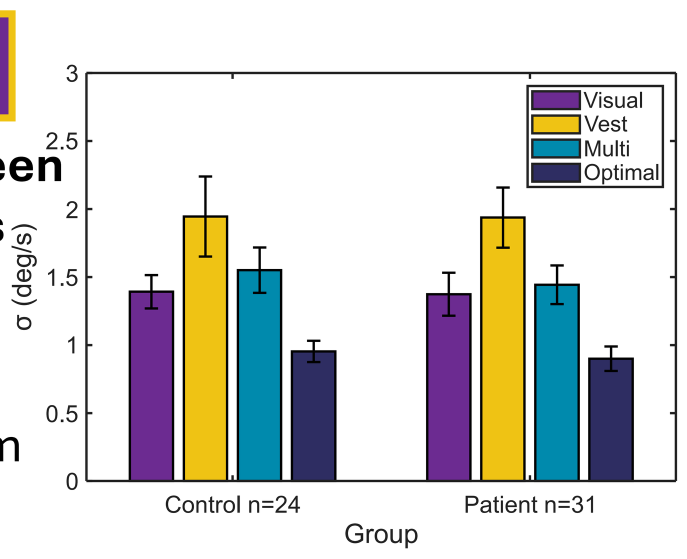


More precise in **multisensory** than either unisensory

Some evidence for near-optimal integration

No clear **differences** between patients and controls

SROM Thresholds



Multisensory **between** unisensory versions

Not good evidence of any integration

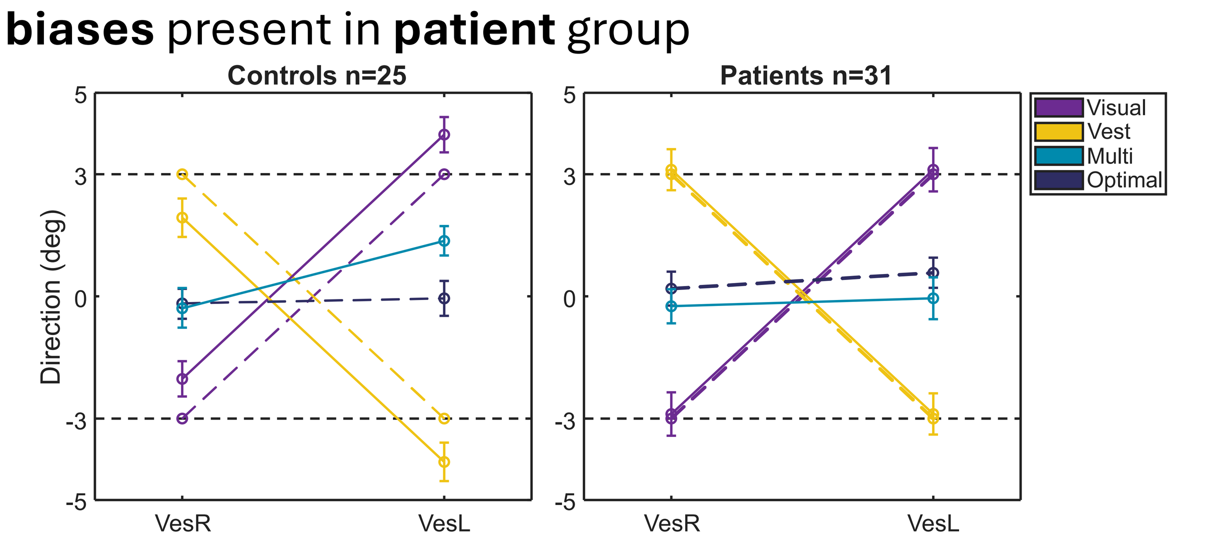
Large deviation from optimal integration

No clear **differences** between patients and controls

Visual Weighting

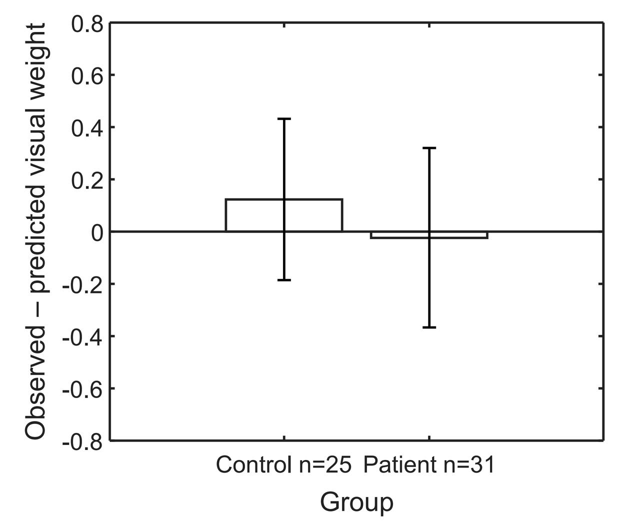
More rightwards bias when the visual stimulus is on the right for the **control group**, could be slight **visual overweighting**

No biases present in **patient** group



No significant visual overweighting by either group, some for controls

Evidence **against** the integration hypothesis



Summary and Conclusions

Persistent Postural Perceptual Dizziness (PPPD) is a functional dizziness disorder which causes a distrust of vestibular information

Multisensory integration involves integrating redundant signals from different sensory systems for the best bias/precision trade-off

Previous research suggests an over-reliance on visual relative to vestibular information

Participants performed multisensory heading and SROM tasks, judging the direction of self- and object- movement

MLE analysis shows similar threshold patterns for controls and patients, with near-optimal integration in the heading task but no evidence for integration in the SROM task

No evidence for overweighting of visual information in patients, contrary to the integration hypothesis

Multisensory integration is intact in PPPD, with appropriate subconscious sensory weighting, differences perhaps at a conscious level instead. This would make sense for a functional disorder

References

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